MULTI-MODAL DEEPFAKE DETECTION VIA SPATIAL-TEMPORAL ENCODING AND CROSS-MODAL FUSION

PHASE II REPORT

Submitted by

RAGHUL J

in partial fulfillment for the award of the degree of

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ANNA UNIVERSITY, CHENNAI

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BONAFIDE CERTIFICATE

Certified that this Thesis titled "**MULTI-MODAL DEEPFAKE DETECTION VIA SPATIAL-TEMPORAL ENCODING AND CROSS-MODAL FUSION**" is the Bonafide work of **RAGHUL J (2023207029)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The rapid evolution of deepfake technology has made detecting forged multimedia content increasingly challenging, especially as manipulations extend beyond visual alterations to include highly realistic audio forgeries. Traditional deepfake detection methods primarily focus on analyzing either video or audio independently, limiting their ability to capture inconsistencies across modalities. This work introduces a multimodal deepfake detection framework that fuses both video and audio modalities to improve detection accuracy and robustness.

The proposed method consists of three key components: (1) Temporal-Spatial Encoder (TSE), (2) Multimodal Joint Decoder, and (3) Cross-Modal Classifier. The TSE extracts discriminative spatial, temporal, and spectral features using Xception, TCN, and Swin Transformer for video and ViT, TCN, and AST for audio. These modality-specific features are then fused in the Multimodal Joint Decoder, which employs binary cross-attention and self-attention mechanisms to align and integrate audio-visual information effectively. Finally, the Cross-Modal Classifier detects inconsistencies between modalities by analyzing fused embeddings and performing real/fake classification using a Transformer based classifier. To enhance feature separation and improve detection robustness, the system is optimized using contrastive loss, cross-entropy loss, and additive angular margin loss. The model is trained and evaluated on the LAV-DF dataset, demonstrating superior performance in both detection accuracy and forgery localization.

சுருக்கம்

டீப்ப்ஃபேக் தொழில்நுட்பத்தின் விரைவு வளர்ச்சி, காணொளி மற்றும் ஒலி போலிகளை உருவாக்குவதன் மூலம், மல்டிமீடியா உள்ளடக்கங்களை கண்டறிதலை பெரும் சவாலாக மாற்றியுள்ளது. பாரம்பரிய முறைகள் காணொளி அல்லது ஒலியை தனித்தனியாக பகுப்பாய்வு செய்யும் நிலையில், மாதிரி இடையே உள்ள முரண்பாடுகளை கண்டறிய முடியவில்லை.

இந்தப் பணி, காணொளி மற்றும் ஒலியை ஒருங்கிணைக்கும் புதிய மல்டிமோடல் டீப்ப்ஃபேக் கண்டறிதல் வடிவமைப்பை முன்வைக்கிறது, இது கண்டறிதல் துல்லியத்தையும் நிலைத்தன்மையையும் மேம்படுத்துகிறது. பரிந்துரைக்கப்பட்ட முறை மூன்று பகுதிகளை கொண்டுள்ளது:

(1) கால-இட குறியீட்டி (TSE),

(2) மல்டிமோடல் கூட்டு டிகோடர்,

(3) கிராஸ்-மோடல் வகைப்படுத்தி.

TSE கூறு காணொளிக்காக Xception, TCN, Swin Transformer மற்றும் ஒலிக்காக ViT, TCN, AST போன்றவற்றைப் பயன்படுத்தி அம்சங்களை வெளிக்கொணர்கிறது. மல்டிமோடல் கூட்டு

டிகோடர், Binary Cross-Attention மற்றும் Self-Attention கொண்டு ஒருங்கிணைப்பு செய்கிறது. பின்னர், கிராஸ்-மோடல் வகைப்படுத்தி ஒருங்கிணைந்த அம்சங்களை பகுப்பாய்ந்து உண்மை/போலி வகைப்படுத்துகிறது.

இந்த முறை contrastive loss, cross-entropy loss மற்றும் additive angular margin loss ஆகியவற்றின் மூலம் பயிற்சி பெறுகிறது. LAV-DF தரவுத்தொகுப்பில் பயிற்சி மற்றும் மதிப்பீட்டின் போது உயர் கண்டறிதல் துல்லியம் மற்றும் போலி பகுதிகளை துல்லியமாக சுட்டிக்காட்டும் திறனை வெளிப்படுத்தியது.

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LIST OF ABBREVIATION

ABBREVIATION	DESCRIPTION
AST	AUDIO SPECTROGRAM TRANSFORMER
CNN	CONVOLUTIONAL NEURAL NETWORK
DFDC	DEEPFAKE DETECTION CHALLENGE
ECL	ENHANCED CONTRASTIVE LEARNING
FFN	FEEDFORWARD NETWORK
LAV-DF	LOCALIZED AUDIO-VIDEO DEEPFAKE
MMD	MULTIMODAL JOINT DECODER
TCN	TEMPORAL CONVOLUTION NETWORK
ViT	VISION TRANSFORMER

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CHAPTER 1

INTRODUCTION

Deepfake technology has advanced significantly in recent years, enabling the creation of highly realistic manipulated videos and audio recordings. These forgeries pose serious threats to digital security, media integrity, and public trust by enabling misinformation, identity fraud, and synthetic impersonation. While deepfake detection methods have evolved to counter these threats, many existing solutions focus solely on either visual or auditory modalities, limiting their effectiveness against sophisticated cross-modal manipulations. The deepfake detection framework, which integrates both video and audio modalities to improve forgery detection and localization accuracy. By leveraging deep learning architectures capable of analyzing spatial, temporal, and spectral inconsistencies, this multimodal approach enhances robustness against evolving deepfake techniques.

1.1 THE NEED FOR MULTIMODAL DEEPFAKE DETECTION

Traditional deepfake detection methods primarily rely on visual cues, such as facial inconsistencies, unnatural blinking, or pixel-level artifacts. However, modern deepfake techniques have become increasingly sophisticated, often preserving high visual fidelity while subtly manipulating audio to create realistic speech synthesis. Such manipulations include voice cloning, lip-syncing, and cross-modal deepfakes, where spoken words and facial expressions are altered in a synchronized manner. These cross-modal inconsistencies are difficult to detect using unimodal approaches, as they exist across both visual and auditory streams. By fusing video and audio information, our proposed framework captures these hidden inconsistencies and improves detection accuracy.

Another major challenge in deepfake detection is temporal inconsistency. Many deepfake algorithms fail to maintain natural temporal transitions, leading to subtle anomalies in lip movement synchronization, speech cadence, and facial dynamics.

Unimodal systems struggle to detect these irregularities as they lack the ability to analyze interdependent features across modalities. Our approach addresses this by employing Temporal-Spatial Encoders (TSE) to extract deep feature representations that account for both spatial and temporal dependencies in video and spectral characteristics in audio.

1.2 OBJECTIVES

The primary objectives of this work are as follows:

- Develop a multimodal detection system that effectively captures inconsistencies across both video and audio modalities.
- Enhance temporal-spatial learning by leveraging deep learning architectures that analyze spatial, temporal, and spectral representations.
- Improve cross-modal fusion mechanisms to align and integrate features from different modalities for better classification.
- Increase robustness against adversarial attacks by utilizing contrastive learning and self-attention mechanisms to detect forged content more effectively.
- Establish a scalable detection framework that generalizes well across diverse deepfake datasets and manipulation techniques.

1.3 PROBLEM STATEMENT

With advancements in deepfake technology, manipulations now extend beyond visual alterations to include highly realistic audio forgeries. Relying solely on visual cues is insufficient, as some deepfakes exhibit near-perfect facial consistency while containing subtle mismatches between speech and lip movements. Existing detection methods fail to capture these cross-modal inconsistencies, making them vulnerable to sophisticated attacks.

To address this, we propose a multimodal approach that fuses audio and video features, leveraging spatial, temporal, and spectral representations. The video encoder captures frame-level features, while the audio encoder extracts speech patterns. A joint decoder with binary cross-attention fuses these features, and a cross-modal classifier detects inconsistencies using a fully connected layer. Optimized with contrastive loss, cross-entropy loss, and additive angular margin loss.

1.4 CHALLENGES

Developing an audio-visual deepfake detection system presents several challenges:

- Deepfake videos are often compressed or intentionally degraded to mimic real-world sharing scenarios, which blurs fine-grained artifacts and makes detection significantly harder, requiring models that are robust to low-resolution and noisy inputs.
- Cross-Modal Feature Misalignment: Aligning video and audio embeddings in a meaningful way remains a complex task due to variations in feature representations.
- Synchronization Issues: Deepfake video often exhibit subtle inconsistencies between speech and lip movements, requiring precise temporal modeling.
- High Computational Complexity: Processing both modalities increases computational demands, necessitating optimized architectures for real-time detection.

1.5 CONTRIBUTION AND IMPACT

The introduction of a multimodal deepfake detection framework represents a significant advancement in combating synthetic media threats. By combining video and audio modalities, the proposed system captures inconsistencies that unimodal models fail to detect, improving generalization against unseen deepfake techniques. This work contributes to the development of more robust, scalable, and real-world applicable deepfake detection solutions, ensuring better security and media authenticity in an era of increasingly sophisticated synthetic content.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 MULTIMODAL DEEPFAKE

Multi-modal deepfake detection relies on capturing audio-visual inconsistencies to improve forgery detection accuracy. AVoiD-DF: Audio-Visual Joint Learning for Detecting Deepfake proposes a framework that integrates a Temporal-Spatial Encoder to embed temporal and spatial information, a Multi-Modal Joint Decoder to fuse multi-modal features and learn inter-modal relationships, and a Cross-Modal Classifier to detect manipulations based on intra-modal and inter-modal disharmony. W Yang et al [1] introduced this approach to address the limitations of unimodal detection methods, which often fail to generalize across diverse forgery techniques. To further enhance evaluation, they developed a benchmark dataset, DefakeAVMiT, which contains manipulated videos where either or both audio-visual modalities may be altered. Experimental results on DefakeAVMiT, FakeAVCeleb, and DFDC demonstrate that AVoiD-DF outperforms existing deepfake detection models, showcasing superior generalization and robustness across various forgery scenarios. However, challenges remain in detecting highly sophisticated forgeries and ensuring the model adapts well to unseen deepfake techniques.

The literature on deepfake detection has evolved from unimodal approaches that analyze either audio or visual content to advanced multi-modal frameworks that address audio-visual inconsistencies and temporal forgeries. Initial visual detection methods utilized convolutional neural networks (CNNs) to capture spatial artifacts such as pixel irregularities, facial landmark distortions, and lighting inconsistencies, while audio-based methods employed spectral and prosodic analysis to identify manipulated speech signals proposed by Miao Liu et al,[2]. However, these unimodal approaches struggled to detect cross-modal anomalies and failed to generalize across diverse datasets. Recent advancements leverage cross-modal

attention mechanisms to detect inter-modal discrepancies, embedding-level fusion techniques with self-attention for temporal dependency modeling, and boundary matching for precise localization of manipulations. Temporal forgery detection, a critical challenge, has been addressed by integrating temporal convolutional networks (TCNs) and multi-dimensional contrastive loss functions to exploit both temporal and inter-modal inconsistencies. This project builds upon these techniques to enhance detection accuracy and localization capabilities for real-world audio-visual deepfakes.

A novel audio-visual fusion and dynamic weighting strategy known as AVT2-DWF is proposed to improve deepfake detection. This technique effectively combines audio and visual cues, recognizing that discrepancies often exist between these modalities in deepfake content. The incorporation of a dynamic weighting mechanism allows the model to adaptively fuse the audio and visual features based on their relevance, enhancing overall detection accuracy. This approach achieves high accuracy while maintaining a simple and efficient architecture, making it particularly suitable for real-time applications where quick decision-making is essential, as introduced by Wang et al. [3]. The flexibility of this strategy enables it to be applied across various media formats, showcasing its potential to enhance multimedia integrity beyond just deepfake detection.

2.1.2 LOW-QUALITY DEEPFAKE

A novel low-quality deepfake detection method leverages a visibility matrix to identify subtle, often unseen artifacts left behind by manipulations. This approach exploits the inherent weaknesses in deepfake generation methods, which can introduce minor inconsistencies that might not be easily noticeable. The method initializes a uniform visibility matrix, which is crucial for enhancing the visibility of imperceptible artifacts present in low-quality deepfake videos. By performing element-wise operations between the visibility matrix and the original dataset, the model creates an enhanced dataset that facilitates the detection of these subtle discrepancies, as developed by Chhabra et al. [4]. This technique not only improves

artifact visibility but also contributes to better generalization across different types of deepfake manipulations, making it a significant advancement in the field.

2.1.3 FRAME-LEVEL DETECTION

In another approach, a fully unsupervised deepfake video detection method employs enhanced contrastive learning (ECL). This innovative technique utilizes a contrastive loss function to learn representations that are invariant to deepfake manipulations, effectively capturing the unique characteristics of both real and fake videos. By integrating convolutional neural networks (CNNs) and recurrent neural networks (RNNs), the method enhances its ability to recognize complex temporal patterns, thereby achieving state-of-the-art performance on several benchmarks. The use of unsupervised learning reduces dependency on labeled datasets, presenting a significant advancement in deepfake detection methodologies, as proposed by Qiao et al. [5]. This method also allows for continuous learning, where the model can adapt to new deepfake techniques without requiring retraining on labeled data.

Li Zhang et al.[6] propose an unsupervised framework for detecting deepfake videos by leveraging the inherent differences in the provenance of real and manipulated video content. The authors argue that while real video frames are captured by cameras, deepfake video frames are typically computer-generated, resulting in distinguishable artifact patterns. Their methodology involves a two-stage clustering process based on Photo-Response Non-Uniformity (PRNU) and noiseprint features.

A robust frame-level detection method employing lightweight Bayesian inference weighting is introduced. This technique focuses on analyzing individual frames to identify deepfake characteristics, achieving high precision in detection. By leveraging Bayesian inference, the model incorporates probabilistic reasoning into its decision-making process, resulting in reliable frame-level analysis. This lightweight approach allows for effective detection of deepfakes without imposing significant computational burdens, making it ideal for real-time applications, as

developed by Zhou et al. [7]. The Bayesian framework enhances the model's ability to quantify uncertainty in its predictions, which is particularly valuable in scenarios where deepfake characteristics may vary widely.

The 4DPM method utilizes a denoising diffusion probabilistic mask for detecting deepfakes in video content. This innovative approach enhances the model's sensitivity to subtle manipulations by applying probabilistic masking techniques that draw attention to discrepancies within video frames. The architecture combines convolutional and recurrent neural networks, facilitating state-of-the-art performance across multiple benchmarks, as proposed by Yang et al. [8]. By focusing on dynamic masking, this method improves the model's ability to filter out noise and irrelevant features, leading to more accurate deepfake detection.

A Motion-magnified 3D residual-in-dense network for deepfake detection is presented. This method employs a 3D residual-in-dense architecture combined with motion magnification techniques to achieve high accuracy in deepfake detection. By focusing on motion characteristics in addition to visual features, this approach enhances the model's detection capabilities, making it effective for challenging deepfake scenarios, as proposed by Mehra et al. [9]. The integration of motion magnification allows for the detection of subtle temporal discrepancies that may otherwise go unnoticed, showcasing the importance of considering motion dynamics in deepfake detection systems.

Guo et al.[10] propose a novel approach by developing a new backbone network structure, the Space-Frequency Interactive Convolution (SFIConv), which efficiently models manipulation clues. The SFIConv incorporates a Multichannel Constrained Separable Convolution (MCSConv) to learn space-frequency features through stages of generation, interaction, and fusion. This innovative method not only improves the accuracy of Deepfake detection but also enhances cross-dataset evaluation by capturing common artifact features.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 SYSTEM ARCHITECHTURE

The proposed work builds upon the foundation established in Phase 1, extending deepfake detection to a multimodal learning framework that analyzes both spatial and temporal aspects of audio and video modalities. Unlike conventional deepfake detection models that focus on only one modality, our system integrates advanced deep-learning architectures to effectively capture cross-modal inconsistencies. By leveraging transformers, convolutional networks, and attention mechanisms, the system is capable of identifying subtle discrepancies between audio and video, thereby enhancing forgery detection accuracy.

The core workflow of the system is divided into two main streams: video processing and audio processing. Both streams initially extract spatial features before passing them through temporal modeling networks to understand the sequential flow. For video, spatial features are extracted using Xception, an advanced CNN architecture known for its ability to capture fine-grained details while maintaining computational efficiency. For audio, the system employs a Vision Transformer (ViT) to process spectrogram representations, treating them as images to extract meaningful frequency-based features. Following the spatial feature extraction, the system moves to temporal modeling.

A Temporal Convolutional Network (TCN) first captures short-term dependencies across both modalities. Then, for video, a Swin Transformer is employed to enhance long-range motion consistency, leveraging a shifted window attention mechanism to process frame transitions more effectively. In the case of audio, a specialized Audio Spectrogram Transformer (AST) refines the features by identifying global spectral variations and anomalies across time.

After processing, the extracted temporal representations from both streams are aligned and fused in a Multimodal Joint Decoder, where cross-attention and self-attention mechanisms play a crucial role in detecting audio-video inconsistencies. This fusion ensures that mismatches in lip movements, speech

patterns, and background sounds are effectively highlighted, enhancing the model's ability to distinguish real from fake content. Finally, the Cross-Modal Classifier performs binary classification based on the fused embeddings, utilizing a combination of contrastive loss, cross-entropy loss, and additive angular margin loss to refine decision-making and ensure high confidence classification.

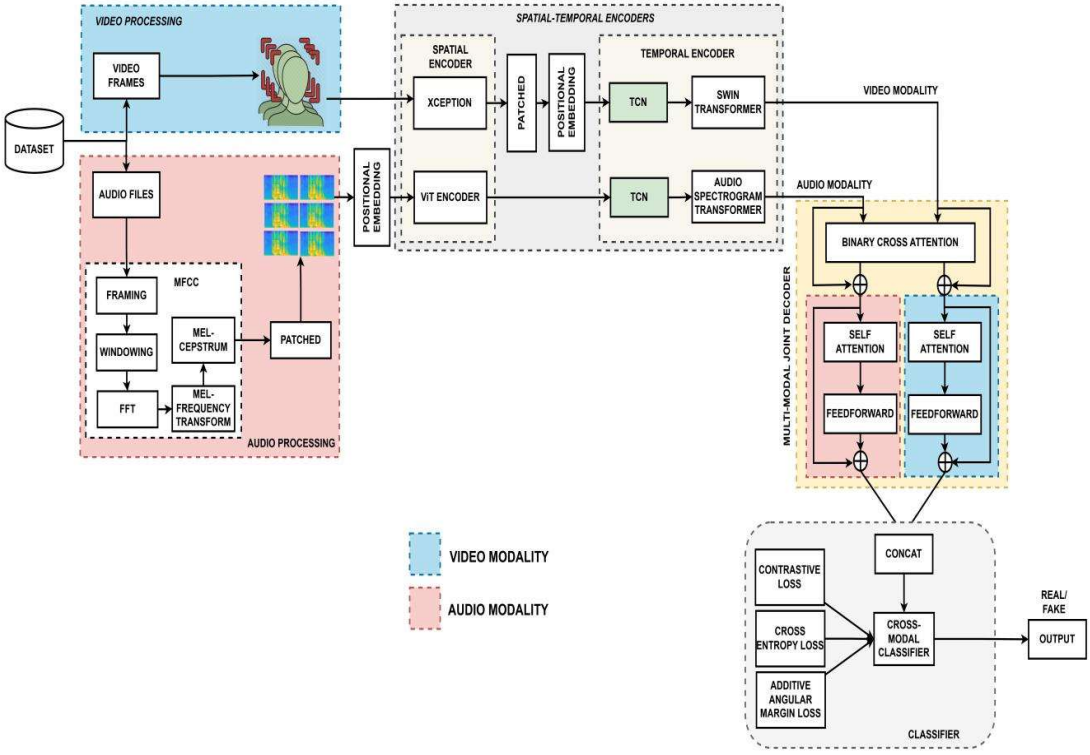


Fig. 3.1 System Architecture

The entire system is designed to jointly learn representations from both modalities, leveraging transformer-based architectures and temporal convolutional techniques to ensure deepfake detection is robust, scalable, and applicable to real-world scenarios. Figure 3.1 illustrates the system architecture, demonstrating how the proposed framework builds upon the multimodal deepfake detection.

The deepfake detection model provides a multimodal framework by integrating both audio and video modalities. The system consists of three key components: the temporal-spatial encoder, the multimodal joint decoder, and the cross-modal classifier. The temporal-spatial encoder processes each modality

separately, extracting spatial features from video using Xception and from audio using a Vision Transformer (ViT) applied to spectrogram representations. These features are then passed through a Temporal Convolutional Network (TCN) for short-term dependencies, while a Swin Transformer enhances long-range motion consistency in video, and an Audio Spectrogram Transformer (AST) refines global spectral variations in audio. The multimodal joint decoder aligns and fuses these temporal representations using cross-attention and self-attention mechanisms to highlight inconsistencies in lip movements, speech patterns, and background sounds. The cross-modal classifier performs binary classification using fused embeddings, employing contrastive loss, cross-entropy loss, and additive angular margin loss to ensure robust decision-making. This phase significantly improves forgery detection by capturing intricate temporal dependencies and modality inconsistencies, making the system highly effective against sophisticated deepfake techniques.

3.2 LIST OF MODULES

The architecture is structured into distinct modules, each responsible for a specific stage of the deepfake detection process, ensuring effective feature extraction, synchronization, and classification. The key modules are as follows:

1. Data Processing & Position Embedding
2. Spatial Encoding & Feature Extraction
3. Temporal Modeling & Synchronization
4. Multimodal Joint Decoder
5. Cross-Modal Classification

3.2.1 DATA PROCESSING & POSITION EMBEDDING

This module begins by processing both audio and video data to generate feature-rich representations for deepfake detection. Video frames are extracted at a consistent frame rate to maintain temporal consistency, then normalized and resized to align with the input requirements of the Xception network. Before that a *visibility matrix v*, which is initialized as a uniform matrix of ones. The dimensions of *v* are

identical to the size of the input image x_i ensure pixel-level correspondence.

Let Preprocessed frames, represent the original dataset D comprising the input preprocessed frames. Using the initial visibility matrix *v*, an enhanced dataset D_{ev} [1] is created by performing element-wise multiplication of *v* with each image in D, expressed as:

$$D_{ev} = \{vx_1, vx_2, vx_3, \ldots vx_m\} \tag{3.1}$$

For audio, waveforms are converted into Mel-Frequency Cepstral Coefficients (MFCCs) and spectrograms, which are further divided into smaller patches to capture detailed acoustic features.

$$\mathbf{S} = \text{LogMelSpec}(x(t)) \in \mathbb{R}^{F \times T} \tag{3.2}$$

Where

F: number of mel frequency bins

T: number of time frames

S: spectrogram matrix

The spectrogram *S* is divided into non-overlapping patches:

$$\mathbf{P}_i = \text{PatchEmbed}(\mathbf{S}) \in \mathbb{R}^{N \times D} \tag{3.3}$$

Where:

N: number of patches

D: embedding dimension for each patch

P_i: embedded patch representation

To enhance learning, position embeddings are applied to both spatial and temporal features, allowing the model to retain structural relationships and sequential order across frames and spectrograms.

$$\mathbf{A}_{\text{input}} = \mathbf{P}_i + \mathbf{E}_{\text{pos}} \tag{3.4}$$

Where A_{input} is the position-aware audio feature input to the next module.

Figure 3.2 depicts the application of position embeddings and their role in preserving structural coherence. This preprocessing step ensures that the features provide a strong foundation for subsequent analysis in the detection pipeline.

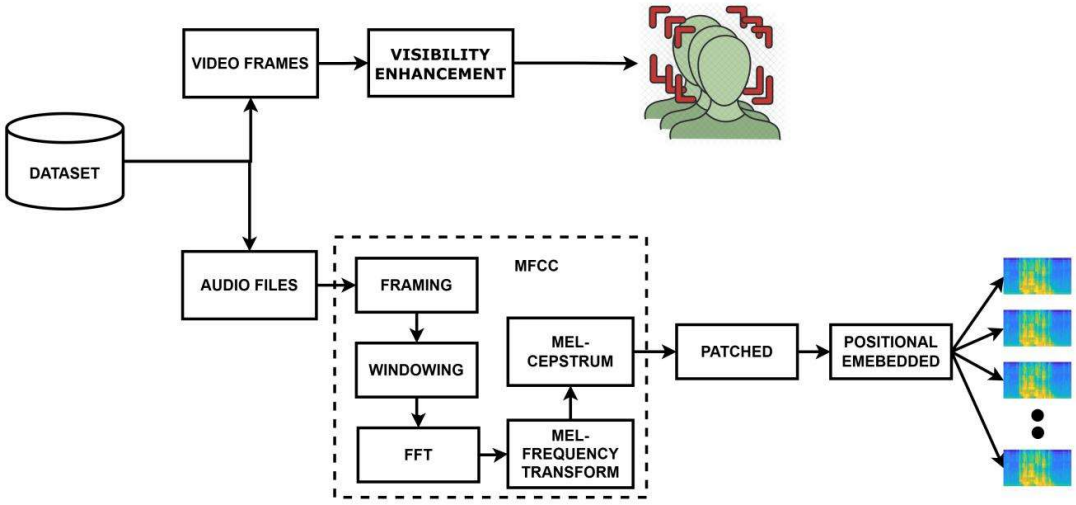


Fig. 3.2 Data Processing & Position Embedding

ALGORITHM: DATA PROCESSING & POSITION EMBEDDING

---

1. video_frames = extract_frames(video_data, frame_rate)

2. audio_mfcc = extract_mfcc(audio_data)

3. spectrograms = transform_audio(audio_mfcc)

4. spectrogram_patches = divide_spectrograms(spectrograms, patch_size)

5. position_embeddings = generate_position_embeddings(spectrogram_patches)

6. frame_embeddings = concatenate(video_frames, position_embeddings)

7. processed_embeddings = linear_layer(frame_embeddings)

8. normalized_embeddings = layer_norm(processed_embeddings)

9. output_embeddings = dropout(normalized_embeddings)

10. return output_embeddings

3.2.2 SPATIAL ENCODING & FEATURE EXTRACTION

This module is responsible for extracting spatial features from both video frames and audio spectrograms before they are processed for temporal modeling. For video, the Xception network is used to capture intricate spatial features while minimizing computational costs through depthwise separable convolutions. This method enhances the model's ability to detect subtle manipulations in deepfakes.

Let each video frame be denoted as I_t , where t ∈ [1,T] (T is the total number of frames). Each frame is passed through a Xception model to extract spatial features:

$$V_t=\text{Xception}(I_t),\ V_t \in R^{dv} \tag{3.5}$$

Stack all frame features over time:

$$V=[V1,V2,\ldots,V_T] \in R^{T\times dv} \tag{3.6}$$

For audio, a Vision Transformer (ViT) processes spectrograms by dividing them into patches and applying self-attention mechanisms. Unlike CNNs, which focus on local features, ViT captures long-range dependencies in the frequency spectrum, making it highly effective in identifying inconsistencies in deepfake audio. From the previous module, we have the positionally embedded patch sequence $A_{input} \in R^{N\times D}$. This sequence is fed into a Vision Transformer (ViT) to extract spatial dependencies between spectrogram patches:

$$A_{vit}=\text{ViT}(A_{input}),A_{vit} \in R^{N\times D} \tag{3.7}$$

Where:

V_t : spatial feature vector from Xception for frame ttt

d_v : dimension of the extracted video feature

A_{vit} : transformer-encoded audio spectrogram patches

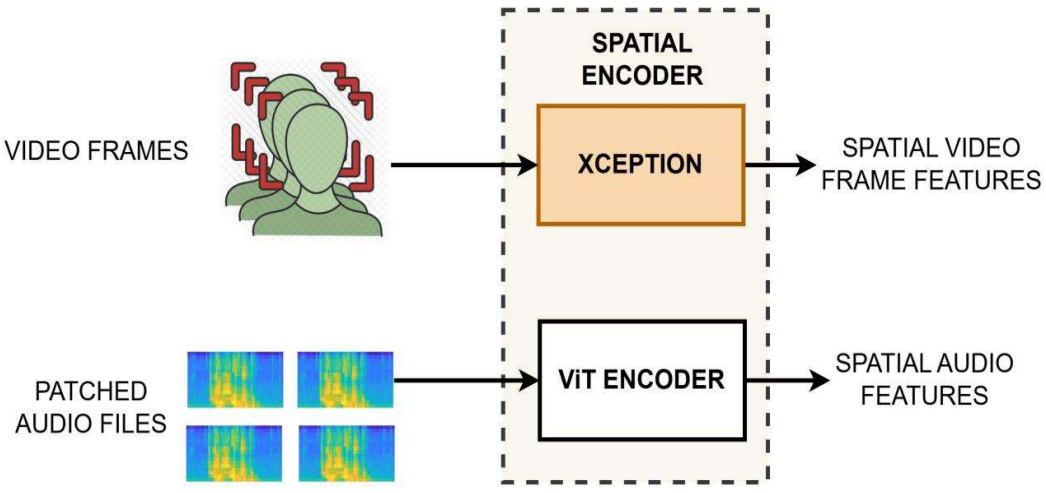


Fig. 3.3 Spatial Encoding & Feature Extraction

Figure 3.3 illustrates the spatial feature extraction process, showing how video and audio features are independently encoded. This representation highlights how self-attention mechanisms in ViT and depthwise convolutions in Xception contribute to robust feature extraction.

ALGORITHM: SPATIAL ENCODING & FEATURE EXTRACTION

---

1. video_spatial_features = xception_network(video_frames)
2. audio_spatial_features = vision_transformer(audio_spectrograms)
3. concatenated_features = concatenate(video_spatial_features, audio_spatial_features)
4. projected_features = linear_layer(concatenated_features)
5. normalized_features = layer_norm(projected_features)
6. spatial_embeddings = dropout(normalized_features)
7. video_spatial_embeddings = spatial_embeddings[:, :video_feature_dim]
8. audio_spatial_embeddings = spatial_embeddings[:, video_feature_dim:]
9. return video_spatial_embeddings, audio_spatial_embeddings
10. End of spatial encoding and feature extraction

At the end of this module, both video and audio data are converted into high-dimensional embeddings, which are then passed to the temporal modeling module for further processing.

3.2.3 TEMPORAL MODELING & SYNCHRONIZATION

After extracting spatial features, this module models the temporal relationships in both video and audio streams, ensuring that both short-term and long-term dependencies are effectively captured. Xception features $V \in R^{T \times dv}$ are divided into patch

$$V_{patch} = \text{PatchEmbed}(V) \in R^{M \times Dv} \quad (3.8)$$

Add learnable positional embeddings:

$$V_{input}=V_{patch}+E_{pos}\in R^{M\times Dv} \tag{3.9}$$

The patched and embedded features are passed into a Temporal Convolutional Network:

$$V_{tcn}=TCN(V_{input})\in R^{M\times Dv} \tag{3.10}$$

For video, a Temporal Convolutional Network (TCN) applies causal convolutions to efficiently learn dependencies between past and future frames, preserving sequential information while reducing computational overhead. The Swin Transformer follows,

$$V_{swin}=SwinTransformer(V_{tcn})\in R^{M\times Dv} \tag{3.11}$$

utilizing shifted window attention to detect inconsistencies such as unnatural facial expressions, abrupt transitions, and artifact movement across frames, which are often key indicators of deepfake manipulation.

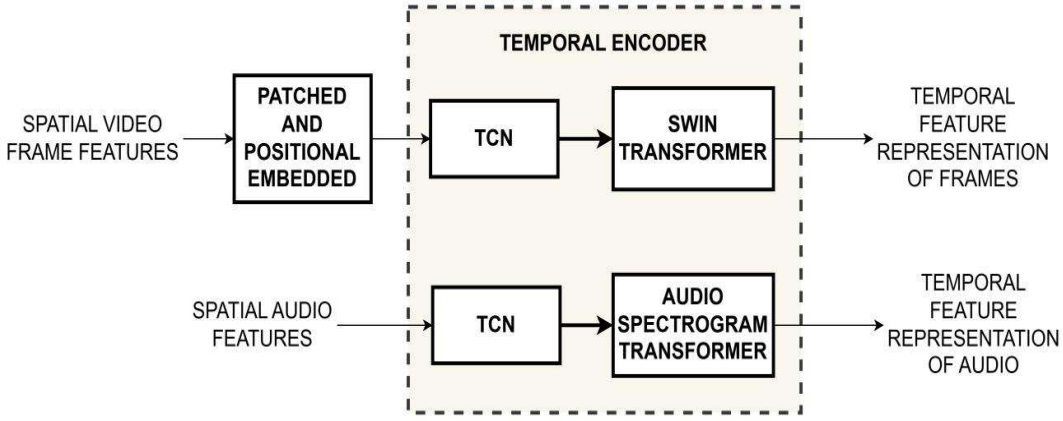


Fig. 3.4 Temporal Modelling & Synchronization

In the audio stream, TCN is used to model short-term dependencies, capturing local variations in speech patterns. The ViT output $A_{vit}\in R^{N\times D}$ is passed through a Temporal Convolutional Network:

$$A_{tcn}=TCN(A_{vit})\in R^{N\times D} \tag{3.12}$$

The temporal features from the TCN are then fed into an Audio Spectrogram Transformer (AST):

$$A_{ast}=AST(A_{tcn})\in R^{N\times D} \quad (3.13)$$

he Audio Spectrogram Transformer (AST) applies self-attention mechanisms to identify global spectral distortions, unnatural pauses, and mismatches between speech and lip movements. This dual approach allows the model to differentiate between natural and manipulated content more effectively.

Figure 3.4 illustrates the temporal modeling process, highlighting how causal convolutions, attention mechanisms, and synchronization steps contribute to robust deepfake detection.

By combining TCN with transformer-based models, this module ensures that temporal features from both modalities are processed comprehensively. The outputs from both streams are then synchronized and aligned for multimodal fusion, setting the stage for cross-modal analysis in the next phase of the pipeline.

ALGORITHM: TEMPORAL MODELING & SYNCHRONIZATION

---

1. vx = temporal_conv_network(video_spatial_embeddings)

2. vx = swin_transformer(vx)

3. ax = temporal_conv_network(audio_spatial_embeddings)

4. ax = audio_spectrogram_transformer(ax)

5. synchronized_features = synchronize(vx, ax)

6. temporal_embeddings = concatenate(synchronized_features, position_embeddings)

7. processed_temporal_embeddings = linear_layer(temporal_embeddings)

8. normalized_temporal_embeddings = layer_norm(processed_temporal_embeddings)

9. output_temporal_embeddings = dropout(normalized_temporal_embeddings)

10. return output_temporal_embeddings

3.2.4 MULTIMODAL JOINT DECODER

This module fuses extracted video and audio features to enhance cross-modal learning. A cross-attention mechanism allows interaction between modalities, refining representations and detecting mismatches in speech and facial expressions. The input features from the video $V_{swin} \in R^{T \times dv}$ are passed through cross-attention layers to allow each modality to attend to the other, capturing dependencies across modalities. Specifically, the audio-guided video attention is given by:

$$V_{att} = \text{CrossAttn}(Q = V_{swin}, K = A_{ast}, V = A_{ast}) \tag{3.14}$$

and the video-guided audio attention is computed as:

$$A_{att} = \text{CrossAttn}(Q = A_{ast}, K = V_{swin}, V = V_{swin}) \tag{3.15}$$

Following this, self-attention layers are applied to further refine the modality-specific features. The self-attended video and audio features are computed as:

$$V' = \text{SelfAttn}(V_{att}) + V_{swin} \tag{3.16}$$

$$A' = \text{SelfAttn}(A_{att}) + A_{ast} \tag{3.17}$$

To dynamically adjust the importance of each modality, a modality-specific weighting α is applied, where $\alpha \in [0,1]$. The fused features are then given by:

$$F_{fused} = \alpha \cdot V' + (1 - \alpha) \cdot A' \tag{3.18}$$

These fused features are then passed through a feedforward network (FFN) to process high-dimensional features and capture inconsistencies:

$$F_{out} = \text{FFN}(F_{fused}) \tag{3.19}$$

The output $F_{out} \in R^{T \times d}$ is now ready for the final classification module, which uses these enhanced multimodal features to detect deepfake manipulations.

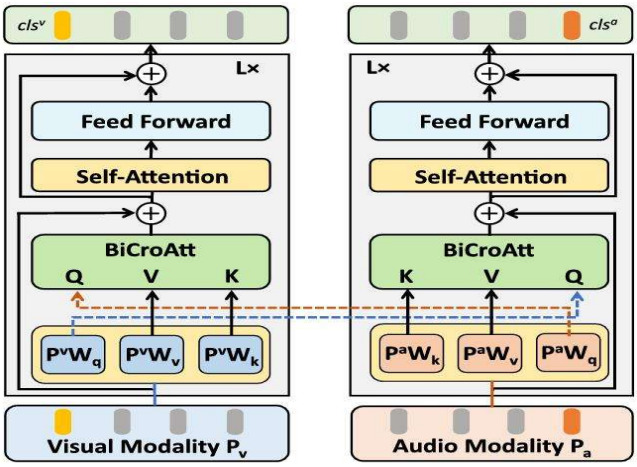


Fig. 3.5 Multimodal Joint Encoder

The attention-based mechanism significantly boosts the model's ability to detect sophisticated deepfake manipulations, improving robustness against adversarial attacks. Figure 3.5 illustrates the multimodal fusion process, showing how cross-attention and self-attention help detect discrepancies, while modality-specific weighting and residual connections preserve key information.

ALGORITHM: MULTIMODAL JOINT DECODER

---

1. fused_features = cross_attention(video_temporal_embeddings, audio_temporal_embeddings)

2. self_attended_features = self_attention(fused_features)

3. feed_forward_features = feed_forward_network(self_attended_features)

4. normalized_features = layer_norm(feed_forward_features)

5. fused_embeddings = dropout(normalized_features)

6. concatenated_embeddings = concatenate(fused_embeddings, spatial_embeddings)

7. projected_embeddings = linear_layer(concatenated_embeddings)

8. normalized_embeddings = layer_norm(projected_embeddings)

9. output_embeddings = dropout(normalized_embeddings)

10. return output_embeddings

3.2.5 CROSS-MODAL CLASSIFIER

The final stage involves classification using a cross-modal classifier that utilizes the fused audio-visual features to determine whether a video is real or fake. The extracted video and audio embeddings are first concatenated into a unified representation, ensuring that both modalities contribute equally to the final decision. This unified representation is then passed through a Transformer model classifier, which learns to distinguish between authentic and manipulated content based on high-dimensional feature relationships.

To optimize classification performance, multiple loss functions are employed. The contrastive loss encourages the model to learn similar representations for augmented views of the same image (positive pairs) and dissimilar representations for different images (negative pairs). It is defined as:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\frac{\mathbf{z}_i\cdot\mathbf{z}_j}{\tau}\right)}{\sum_{k=1}^{2N}1_{[k\neq i]}\exp\left(\frac{\mathbf{z}_i\cdot\mathbf{z}_k}{\tau}\right)} \tag{3.20}$$

Where:

z_i, z_j are feature vectors of positive pairs

τ is the temperature parameter.

$1_{[k\neq i]}$ excludes the anchor image from the denominator.

Cross-entropy loss directly enhances classification accuracy by minimizing the difference between predicted and actual labels. Additionally, additive angular margin loss improves the decision boundary by enforcing a larger separation between real and fake feature vectors, making it easier to distinguish between subtle deepfake manipulations.

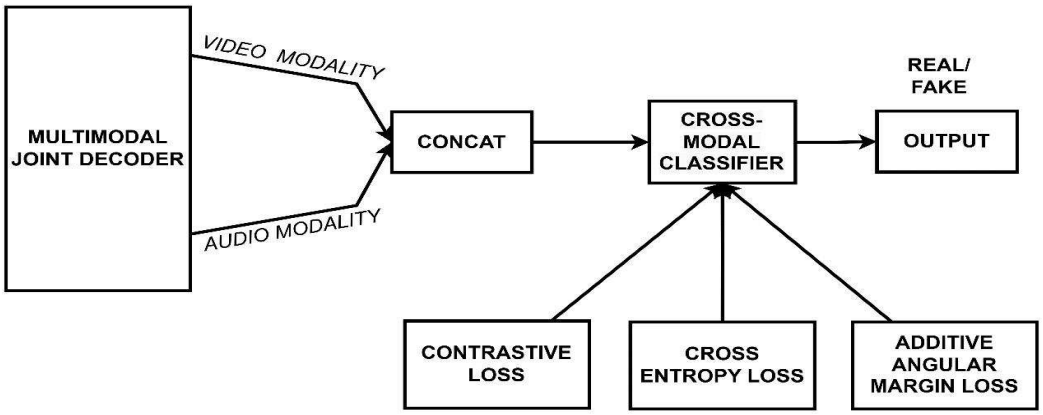


Fig. 3.6 Cross-Modal Classification

Figure 3.6 illustrates the classification process, highlighting how the fused embeddings are transformed through the classifier and optimized using multiple loss functions.

ALGORITHM: CROSS-MODAL CLASSIFIER

---

1. unified_representation = concatenate(video_embeddings, audio_embeddings)

2. class_logits = fully_connected_layer(unified_representation)

3. contrastive_loss = contrastive_loss(class_logits, true_labels)

4. cross_entropy_loss = cross_entropy_loss(class_logits, true_labels)

5. additive_angular_margin_loss = additive_angular_margin_loss(class_logits, true_labels)

6. total_loss = contrastive_loss + cross_entropy_loss + additive_angular_margin_loss

7. optimizer = AdamOptimizer(learning_rate=0.001)

8. optimizer.minimize(total_loss)

9. predicted_class = argmax(class_logits)

10. return predicted_class

The integration of these loss functions ensures that the classifier remains highly accurate, even when dealing with highly sophisticated deepfake manipulations. Dropout and batch normalization techniques are incorporated to improve generalization and prevent overfitting, making the model robust across different datasets and real-world scenarios.

After the final Transformer layer:

$$\hat{y}=\sigma(W\cdot H^{(L)}+b) \tag{3.21}$$

Where:

- $\hat{y} \in [0,1]$ is the predicted probability of the video being fake
- $W \in R^{1\times d}$, $b \in R$ are learnable parameters
- σ is the sigmoid activation function

Feature normalization ensures well-distributed embeddings, reducing bias in classification. A confidence-based thresholding mechanism refines predictions by filtering out uncertain classifications. The model also employs adaptive learning rates for efficient convergence and improved generalization. By leveraging these techniques, the classifier remains effective in detecting deepfakes across diverse and challenging scenarios.

CHAPTER 4

IMPLEMENTATION DETAILS

4.1 EXPERIMENTAL RESUTS

DATA PROCESSING & POSITIONAL EMBEDDING

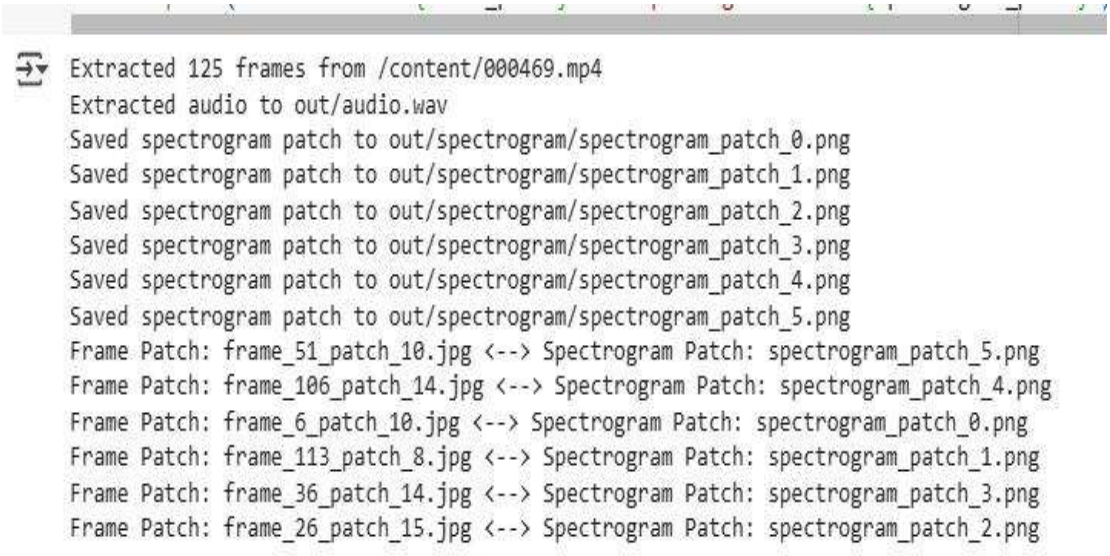


Fig. 4.1 Data extraction and storing

The implementation of our deepfake detection framework begins with extracting both video frames and audio from input videos, transforming them into fixed-size patches, and applying positional embeddings to retain spatial and temporal structures. The video processing pipeline involves extracting frames at a fixed frame rate (e.g., 30 FPS), converting them to grayscale, and dividing each frame into non-overlapping patches of size 32×32 pixels. To preserve the spatial structure of these patches, we apply sinusoidal positional encoding, assigning unique position-aware values to each patch. Figure 4.1 confirms the successful data extraction and storage, ensuring that the frame patches are properly saved for further processing.

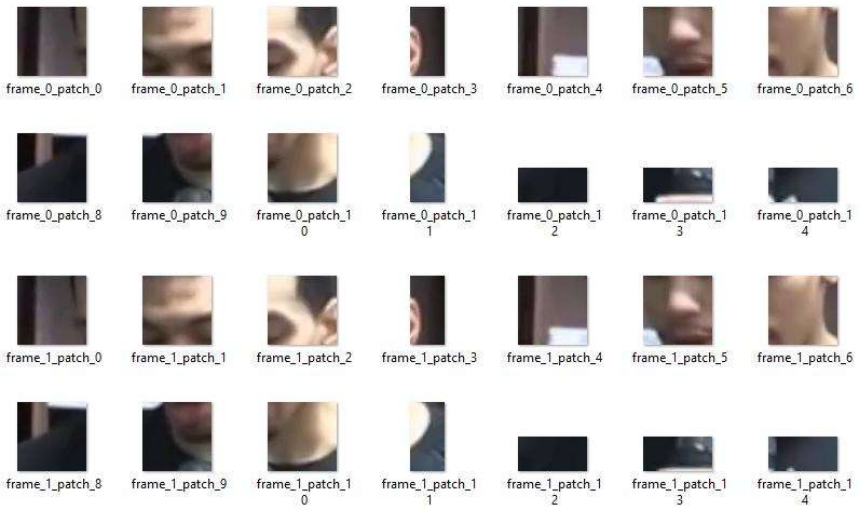


Fig. 4.2 Frame patches

Next, the extracted video frames are processed by visualizing individual patches, which help in analyzing how each region contributes to deepfake detection. These patches allow the model to focus on localized inconsistencies within video frames, making it more effective in identifying deepfake artifacts. Figure 4.2 illustrates the visualized frame patches, demonstrating how the original frames are divided into smaller regions for enhanced feature extraction.

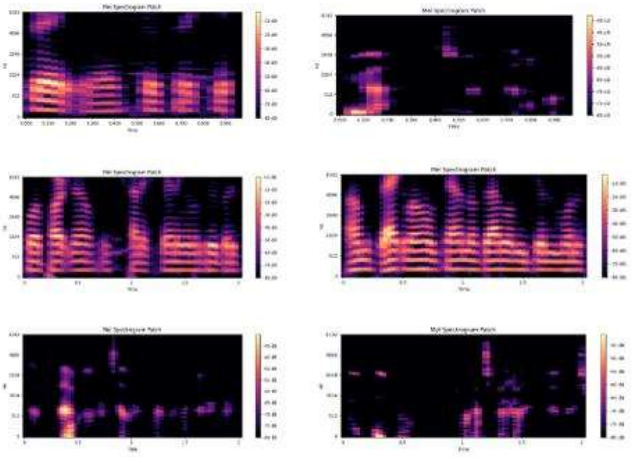


Fig. 4.3 Audio Spectrogram Patches

Simultaneously, the audio stream undergoes preprocessing, where it is resampled to 16 kHz and converted into Mel-Frequency Cepstral Coefficients (MFCCs). These MFCC representations capture essential frequency-time features

of the audio signal, which are further divided into 32×32 patches to retain localized spectral variations. Positional embeddings are applied to these patches to introduce temporal sequence awareness, enabling the model to detect inconsistencies in deepfake speech. Fig 4.3 presents the audio spectral patches, showcasing the segmented MFCC representations utilized in cross-modal deepfake detection.

SPATIAL ENCODING AND FEATURE EXTRACTION

(i) Xception

```
# Train model
train_xception(frames_root, model_save_path)
```

```
C:\Users\Raghul J\AppData\Local\Programs\Python\Python311\Lib\site
ion to current legacy_xception.
  model = create_fn(
Epoch [1/10] | Loss: 1.5558 | Accuracy: 0.9991
Epoch [2/10] | Loss: 0.0052 | Accuracy: 1.0000
Epoch [3/10] | Loss: 0.0025 | Accuracy: 1.0000
Epoch [4/10] | Loss: 0.0014 | Accuracy: 1.0000
Epoch [5/10] | Loss: 0.0009 | Accuracy: 1.0000
Epoch [6/10] | Loss: 0.0006 | Accuracy: 1.0000
Epoch [7/10] | Loss: 0.0004 | Accuracy: 1.0000
Epoch [8/10] | Loss: 0.0002 | Accuracy: 1.0000
Epoch [9/10] | Loss: 0.0002 | Accuracy: 1.0000
Epoch [10/10] | Loss: 0.0001 | Accuracy: 1.0000
Model saved at C:\Users\Raghul J\Documents\phase 2\xception.pth
```

Fig. 4.4 Training the Xception model

The Xception model is trained to extract spatial features from video frames, enabling effective identification of deepfake artifacts. The input to the Xception model consists of preprocessed video frames extracted from the given video dataset. Each frame is resized and normalized before being passed through the convolutional layers of the Xception network. The final trained Xception model is then saved and used for feature extraction in the subsequent phase. Figure 4.4 illustrates the training process of the Xception model.

```
# Extract and Save Features in Video-Named Folders
extract_xception_features(frames_root, model_save_path, output_dir)
```

```
Saved features for 000811 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000811
Saved features for 000812 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000812
Saved features for 000813 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000813
Saved features for 000814 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000814
Saved features for 000815 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000815
Saved features for 000816 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000816
Saved features for 000817 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000817
Saved features for 000818 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000818
Saved features for 000819 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000819
Saved features for 000820 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000820
Saved features for 000821 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000821
Saved features for 000822 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000822
Saved features for 000823 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000823
Saved features for 000824 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000824
Saved features for 000825 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000825
Saved features for 000826 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000826
Saved features for 000827 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000827
Saved features for 000828 in C:\Users\Raghul J\Documents\phase 2\xception_features2\000828
```

Fig. 4.5 Xception - Output features

Once the Xception model is trained, it is used as a feature extractor to generate spatial feature representations from video frames. The input to this stage consists of video frames that have undergone preprocessing, including frame extraction, resizing, and normalization. The trained Xception model processes each frame and outputs a feature map representing spatial characteristics. These extracted feature maps are then saved for further processing.

Figure 4.5 illustrates the process of extracting and saving features using the trained Xception model. The saved features serve as input to the next stage, where a Temporal Convolutional Network (TCN) is employed to model temporal dependencies between frames for improved deepfake detection accuracy.

(ii) Vision Transformer

The ViT model is trained to extract spatial features from audio spectrogram patches, allowing it to capture important frequency and time-domain patterns that help distinguish between real and fake audio depicts in Fig.4.6. The input to the ViT model consists of preprocessed audio representations, where raw

audio signals are first converted into spectrograms. These spectrograms are then patched into smaller segments before being passed to the ViT model.

```
model_save_path = "audio_vit_features.pth"
train_vit_audio(data_dir, model_save_path)
```

```
Epoch [1/10], Loss: 0.17569214602311453
Epoch [2/10], Loss: 0.038417983489731945
Epoch [3/10], Loss: 0.013019879426186284
Epoch [4/10], Loss: 0.007624791158984105
Epoch [5/10], Loss: 0.0035569145887469253
Epoch [6/10], Loss: 0.0021763056089791157
Epoch [7/10], Loss: 0.001365890047357728
Epoch [8/10], Loss: 0.0007270132385504743
Epoch [9/10], Loss: 0.0006211996709074205
Epoch [10/10], Loss: 0.0003644421764571841
Model saved to audio_vit_features.pth
```

Fig. 4.6 ViT training

The ViT model applies self-attention mechanisms to learn hierarchical feature representations from the patched spectrograms. During training, the model optimizes a classification loss function to differentiate between authentic and manipulated audio. The final trained ViT model is saved and later used as a feature extractor.

```
Extracted Features Shape: (1, 128)
[[-0.01928449  0.0282811  -0.02997559 -0.00521409  0.01476807 -0.04485905
   0.01974162 -0.03431411  0.07678259 -0.05731553  0.00166584  0.02757292
   0.00722858 -0.03478616 -0.03229161  0.02272964  0.04354717 -0.01374489
   0.02255877  0.05784248 -0.03881178 -0.00728974  0.03831171  0.00934963
  -0.05509534  0.00243503  0.04394311 -0.00918421  0.00236129 -0.07259111
  -0.07612091 -0.05517542 -0.01111935 -0.04496116  0.01653874  0.02921493
   0.03787293  0.00482878 -0.03205018  0.01897644 -0.02305172 -0.00267216
   0.03017785 -0.03252137 -0.02854312 -0.02558805  0.02786718 -0.00368296
   0.01525495 -0.01771064 -0.03405666 -0.02309883  0.03058442 -0.00389354
  -0.0007585   0.00451353  0.02533169 -0.00760107 -0.01247539 -0.01137535
  -0.02319441  0.00935875 -0.02772233 -0.022038    0.03616875 -0.01879089
   0.02263443 -0.04399232  0.01307351 -0.01187376 -0.00817858  0.01640036
  -0.02182916  0.03787368 -0.01155039  0.00130985  0.01513279 -0.00974456
  -0.01811789  0.00927247 -0.01232887  0.01765975 -0.00859543 -0.02218167
  -0.02788093  0.01305282  0.01980918  0.04599681 -0.01109411 -0.05743704
   0.02211945  0.00518006  0.10434777 -0.02486271 -0.00823703  0.00582759
   0.02743852 -0.03003243  0.04530026 -0.03062361 -0.03286744 -0.00074157
   0.0151305   0.00392244  0.01651117  0.02986022  0.01659186 -0.01759662
  -0.02054256 -0.05168229  0.03795019  0.04480346 -0.02804558  0.0213531
  -0.00322395 -0.02655891  0.01086707 -0.0156465  -0.03602929  0.03322691
  -0.01451331 -0.01002459 -0.0657994  -0.04846137 -0.04305555 -0.04605751
  -0.02609298 -0.02409226]]
```

Fig. 4.7 ViT - Output features

Once the ViT model is trained, it is used as a feature extractor to generate spatial feature representations from patched spectrograms. The input to this stage consists of preprocessed and patched spectrograms, where each spectrogram is divided into fixed-size patches before being passed through the trained ViT model.

The ViT processes the patches through multiple self-attention layers, producing a feature embedding for each patch. These extracted feature embeddings are then enhanced with positional encoding to retain the order and structure of the original spectrogram. The final transformed features are saved for the next stage of the pipeline as shown in Figure 4.7.

Similar to the video processing pipeline, the extracted audio features are then passed through a Temporal Convolutional Network (TCN) to model the temporal dependencies across time. This allows the system to analyze both spatial and temporal characteristics of the audio, improving the deepfake detection performance.

TEMPORAL MODELING AND SYNCHRONIZATION
(i) TCN - Video

```
train_tcn(features_dir, model_save_path)
```

```
Epoch [1/10], Loss: 0.2768
Epoch [2/10], Loss: 0.4016
Epoch [3/10], Loss: 0.2364
Epoch [4/10], Loss: 0.1484
Epoch [5/10], Loss: 0.2990
Epoch [6/10], Loss: 0.1699
Epoch [7/10], Loss: 0.1578
Epoch [8/10], Loss: 0.1526
Epoch [9/10], Loss: 0.1448
Epoch [10/10], Loss: 0.1322
Model saved at tcn_video_model.pth
```

Fig. 4.8 TCN training

The input to the TCN model consists of patched and positionally embedded features extracted from the Xception encoder. These features represent the spatial characteristics of video frames and have undergone preprocessing steps, including frame extraction, patching, and positional embedding. The patched features capture local spatial patterns, while positional embeddings ensure that the model retains information about the relative position of each patch. The training of the model is shown in Figure 4.8.

The TCN is designed to model temporal relationships between consecutive frames. It employs causal convolutions and dilated convolutions to capture long-range dependencies in the sequence without requiring recurrent connections. The model consists of multiple convolutional layers, each followed by activation functions and residual connections to ensure efficient gradient flow and prevent vanishing gradients. The output from the TCN is a high-level temporal feature representation that encodes sequential dependencies across frames.

```
extract_tcn_features(features_dir, model_path, output_dir)
```

```
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0000.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0001.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0002.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0003.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0004.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0005.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0006.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0007.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0008.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0009.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0010.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0011.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0012.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0013.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0014.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0015.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0016.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0017.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0018.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0019.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0020.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0021.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0022.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0023.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0024.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0025.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0026.npy
Saved TCN feature: C:\Users\Raghul J\Documents\phase 2\tcn_VIDEO_features\000514\frame_0027.npy
```

Fig. 4.9 TCN - Output features

After training, the TCN model is used to process video sequences by taking the patched and embedded features as input and generating temporally-aware feature representations as shown in Figure 4.9. These representations are essential for distinguishing between authentic and manipulated videos, as deepfake artifacts often exhibit inconsistencies over time.

The extracted TCN features are then passed to a Cross-Modal Classifier, which integrates information from both the video and audio streams. This multimodal fusion approach enhances the detection capability, making the system robust against different types of deepfake manipulations. The trained TCN model is saved and utilized in the final classification stage to determine the authenticity of input videos.

(ii) Swin Transformer - Video

```
Final input shape to Transformer: torch.Size([10, 1, 768])
Epoch [9/10], Avg Loss: 0.0278
Before embedding: torch.Size([16, 1, 512])
After embedding: torch.Size([16, 1, 768])
Final input shape to Transformer: torch.Size([16, 1, 768])
Before embedding: torch.Size([16, 1, 512])
After embedding: torch.Size([16, 1, 768])
Final input shape to Transformer: torch.Size([16, 1, 768])
Before embedding: torch.Size([10, 1, 512])
After embedding: torch.Size([10, 1, 768])
Final input shape to Transformer: torch.Size([10, 1, 768])
Epoch [10/10], Avg Loss: 0.0263
Model saved to C:/Users/Raghul J/Documents/phase 2/swin_transformer.pth
```

Fig. 4.10 Swin Transformer training

The input to the Swin Transformer consists of temporally refined features generated by the TCN. These features have already undergone spatial feature extraction using the Xception model, patching, positional embedding, and temporal

modeling through the TCN. The Swin Transformer processes these representations to learn hierarchical spatial-temporal dependencies within the video frames.

Unlike traditional transformers, which operate on full-frame token sequences, the Swin Transformer employs a shifted window mechanism, enabling localized attention computation while maintaining global context through window shifting across layers. This hierarchical structure allows the model to capture both fine-grained details and long-range dependencies, making it particularly effective for analyzing subtle deepfake artifacts distributed across frames. The training and saving of model is shown in Figure 4.10.

The model consists of multiple Swin Transformer blocks, each comprising self-attention, feed-forward layers, and residual connections to facilitate stable training. The network progressively refines the extracted temporal features, producing a robust spatial-temporal feature representation for deepfake classification.

```
# Extract features
extract_swin_features(feature_root, model_path, output_root)
```

```
Processed: frame_0000.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0000.npy
Processed: frame_0001.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0001.npy
Processed: frame_0002.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0002.npy
Processed: frame_0003.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0003.npy
Processed: frame_0004.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0004.npy
Processed: frame_0005.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0005.npy
Processed: frame_0006.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0006.npy
Processed: frame_0007.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0007.npy
Processed: frame_0008.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0008.npy
Processed: frame_0009.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0009.npy
Processed: frame_0010.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0010.npy
Processed: frame_0011.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0011.npy
Processed: frame_0012.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0012.npy
Processed: frame_0013.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0013.npy
Processed: frame_0014.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0014.npy
Processed: frame_0015.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0015.npy
Processed: frame_0016.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0016.npy
Processed: frame_0017.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0017.npy
Processed: frame_0018.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0018.npy
Processed: frame_0019.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0019.npy
Processed: frame_0020.npy → C:/Users/Raghul J/Documents/phase 2/swin_VIDEO_features\000514\frame_0020.npy
```

Fig. 4.11 Swin Transformer - Output features

Once trained, the Swin Transformer is utilized to process video sequences, generating enhanced feature representations that incorporate both spatial characteristics and temporal dependencies. These refined features are subsequently passed to a Multimodal Joint Decoder, which integrates information from both audio and video streams.

The decoder fuses the extracted video-based Swin Transformer features with audio-based AST (Audio Spectrogram Transformer) features, ensuring that inconsistencies in both modalities are effectively captured depicts in Figure 4.11. Finally, the Cross-Modal Classifier takes the joint representation as input and produces a binary classification output, determining whether the given video is authentic or deepfake.

(iii) TCN - Audio

```
    model_save_path = "audio_tcn_model.pth"
    train_audio_tcn(features_root_dir, model_save_path)
```

```
Epoch [1/10], Loss: 0.16518724958101907
Epoch [2/10], Loss: 0.041550999507308006
Epoch [3/10], Loss: 0.023995128460228443
Epoch [4/10], Loss: 0.017255014119048912
Epoch [5/10], Loss: 0.014405584273238977
Epoch [6/10], Loss: 0.008202117479716739
Epoch [7/10], Loss: 0.005745831799382965
Epoch [8/10], Loss: 0.005158897722139955
Epoch [9/10], Loss: 0.004806837532669306
Epoch [10/10], Loss: 0.004135577784230311
Model saved to audio_tcn_model.pth
```

Fig. 4.12 TCN – Audio Training

The input to the TCN consists of patched and positionally embedded features derived from the ViT-based audio feature extractor. These features represent local spectro-temporal patterns from audio spectrograms, which are then temporally modeled using the TCN architecture.

The TCN consists of stacked 1D dilated causal convolutional layers, enabling it to effectively capture long-term dependencies while preserving the sequential order of audio data. Unlike recurrent networks, which suffer from vanishing gradient issues, the TCN leverages dilated convolutions to expand its receptive field exponentially, ensuring that it can analyze both short-term and long-term temporal dependencies in audio signals. The model is trained and saved as shown in Figure 4.12.

```
Saved TCN features for 000514
Saved TCN features for 000515
Saved TCN features for 000516
Saved TCN features for 000517
Saved TCN features for 000518
Saved TCN features for 000519
Saved TCN features for 000520
Saved TCN features for 000521
Saved TCN features for 000522
Saved TCN features for 000523
Saved TCN features for 000524
Saved TCN features for 000525
Saved TCN features for 000526
Saved TCN features for 000527
Saved TCN features for 000528
Saved TCN features for 000529
Saved TCN features for 000530
Saved TCN features for 000531
Saved TCN features for 000532
Saved TCN features for 000533
Saved TCN features for 000534
Saved TCN features for 000535
Saved TCN features for 000536
```

Fig. 4.13 Saved features from Audio Patches

Once trained, the TCN processes audio sequences, transforming them into temporally refined representations that capture the nuanced variations in speech depicts in Figure 4.13. These enhanced features are then passed to the AST (Audio Spectrogram Transformer) for further processing.

The AST module employs self-attention mechanisms to capture global dependencies across the audio spectrum, complementing the local dependencies extracted by the TCN. This combination ensures that both fine-grained variations

and global acoustic patterns are preserved, allowing the model to detect inconsistencies introduced by deepfake audio synthesis techniques.

The final processed audio features are then integrated with the video-based Swin Transformer features in the Multimodal Joint Decoder, leading to a comprehensive multimodal deepfake detection pipeline.

(iv) Audio Spectrogram Transformer

```
    tcn_features_dir = r"C:\Users\Raghul J\Docum
    ast_model_save_path = "ast_audio_model.pth"

    train_ast(tcn_features_dir, ast_model_save_p
```

```
Epoch [1/10], Loss: 0.14262904350956282
Epoch [2/10], Loss: 0.029576247557997704
Epoch [3/10], Loss: 0.02124787463496129
Epoch [4/10], Loss: 0.01812579855322838
Epoch [5/10], Loss: 0.01596100628376007
Epoch [6/10], Loss: 0.013788184616714716
Epoch [7/10], Loss: 0.01308555481955409
Epoch [8/10], Loss: 0.01234743914877375
Epoch [9/10], Loss: 0.011924237478524446
Epoch [10/10], Loss: 0.011268696747720242
Model saved to ast_audio_model.pth
```

Fig. 4.14 AST training

The input to the AST consists of temporally refined features generated by the TCN. These features encapsulate both short-term phonetic variations and long-range speech dependencies, forming a structured representation of the temporal and spectral properties of the audio signal.

AST is built on the Transformer architecture, where multi-headed self-attention layers are used to model relationships between different audio segments. This allows the network to capture global contextual cues while preserving local

variations. Unlike CNN-based approaches, AST does not rely on fixed receptive fields but instead learns contextual relationships dynamically.

During training as shown in Figure 4.14, the AST model processes patched and embedded audio features using a hierarchical attention mechanism. The positional embeddings introduced earlier in the pipeline enable AST to understand the sequential nature of the input. Layer normalization, dropout, and self-attention layers are used to stabilize training and improve generalization.

```
extract_and_save_ast_features(ast_model_path, tcn_features_dir, ast_features_dir)
```

```
Saved AST features for 000514
Saved AST features for 000515
Saved AST features for 000516
Saved AST features for 000517
Saved AST features for 000518
Saved AST features for 000519
Saved AST features for 000520
Saved AST features for 000521
Saved AST features for 000522
Saved AST features for 000523
Saved AST features for 000524
Saved AST features for 000525
Saved AST features for 000526
Saved AST features for 000527
```

Fig. 4.15 AST features extracted and saved

Once trained, the AST transforms the TCN-refined audio representations into high-dimensional embeddings that capture global temporal dependencies and frequency-based variations in speech. These embeddings are crucial for detecting deepfake-induced artifacts, such as unnatural prosody, abrupt pitch shifts, and synthetic distortions. The extracted features are stored in corresponding video named folders as shown in figure 4.15.

The final output of AST consists of contextualized audio embeddings, which are then passed to the Multimodal Joint Decoder. Here, the AST-extracted features are combined with Swin Transformer-based video features to achieve comprehensive multimodal deepfake detection. By integrating audio and video cues,

the system effectively differentiates between genuine and manipulated media, even in low-quality and adversarially perturbed inputs.

MULTIMODAL JOINT DECODER

The Multimodal Joint Decoder (MMD), which is designed to fuse and refine temporally encoded features from both the audio and video pipelines. The video features extracted via the Swin Transformer and the audio features from the AST module are used as inputs to the MMD.

```
Epoch 1/10 - Loss: 1.0851
Epoch 2/10 - Loss: 0.0120
Epoch 3/10 - Loss: 0.0220
Epoch 4/10 - Loss: 0.0228
Epoch 5/10 - Loss: 0.0165
Epoch 6/10 - Loss: 0.0165
Epoch 7/10 - Loss: 0.0122
Epoch 8/10 - Loss: 0.0122
Epoch 9/10 - Loss: 0.0094
Epoch 10/10 - Loss: 0.0088
Multimodal Decoder saved to: models\multimodal_decoder.pth
```

Fig. 4.16 Multimodal Joint Decoder training

The decoder employs a cross-attention mechanism to facilitate interaction between modalities, allowing audio to guide the interpretation of visual content and vice versa. This helps identify mismatches between lip movements and speech patterns, a common indicator in deepfake videos. Following this, self-attention layers are applied independently to each modality to further refine the learned representations. The refined embeddings are then passed through a feedforward network (FFN), followed by modality-specific weighting and residual connections, ensuring preservation of modality-specific features while allowing for enhanced fusion. The training process of the MMD, including the application of cross-modal attention and gradient flow, is illustrated in figure 4.16.

```
[[-4.74033691e-02  5.88610917e-02  4.71702963e-02  2.75626965e-02
  -6.23870417e-02 -2.70190053e-02  3.34531218e-02 -5.73658943e-02
  -7.89721832e-02  7.12889284e-02 -7.57331252e-02  2.33275145e-02
  -4.81455810e-02 -3.27865630e-02  7.05010593e-02  5.39041646e-02
   5.91404177e-02 -4.73095030e-02  4.50818762e-02 -9.43003595e-02
  -2.25533303e-02 -6.06087595e-02  6.62879795e-02  1.20345969e-03
  -6.58500195e-02  7.45192915e-02 -6.26537055e-02 -8.06385279e-02
   6.16842732e-02 -5.08280750e-03 -8.81126076e-02  8.23210180e-02
  -5.58843240e-02 -6.55194372e-02  3.34247686e-02  4.44643721e-02
  -4.66982201e-02 -5.36613837e-02  3.25642899e-02  7.89299607e-02
  -3.48369498e-03  4.74446379e-02 -6.86471835e-02  5.43409809e-02
   5.32788634e-02  5.19875139e-02 -9.97933149e-02 -7.98769817e-02
   6.11978769e-02  4.06455249e-02  2.86587887e-03  5.13893142e-02
   7.56312832e-02  4.72668149e-02  6.53223097e-02  7.28618801e-02
  -8.68381634e-02 -3.25848423e-02 -1.01158708e-01 -3.62614691e-02
   5.15725762e-02 -1.27007961e-01  3.58524285e-02  8.06435291e-03
  -6.97601587e-02  8.64889249e-02 -1.07353933e-01 -4.10999097e-02
  -1.03022344e-03 -7.82822073e-02 -9.10672918e-02  5.56629412e-02
```

Fig. 4.17 Extracted features from MMD

Once trained, the MMD generates highly discriminative fused features capable of capturing fine-grained inconsistencies across modalities. An example output of the joint decoder, showing how audio and visual embeddings are aligned or diverge across time, is visualized in figure 4.17, highlighting its effectiveness in detecting subtle deepfake manipulations.

CROSS-MODAL CLASSIFIER

The final component of the deepfake detection pipeline is the cross-modal classifier, which plays a key role in determining whether the input video is real or fake. This stage begins by taking the fused audio and video features generated by the multimodal joint decoder. These features contain rich information from both modalities, including speech patterns and facial expressions that have already been aligned and refined. The first step is to combine these features into a unified representation that captures both the temporal and semantic alignment between audio and video. This fusion process is illustrated in figure 4.18, where the temporally aligned features from both streams are merged into a joint feature vector.

```
[✓] Fused features saved for 000469
[✓] Fused features saved for 000537
[✓] Fused features saved for 000571
[✓] Fused features saved for 000588
[✓] Fused features saved for 014779
[✓] Fused features saved for 025846
[✓] Fused features saved for 025891
[✓] Fused features saved for 025923
[✓] Fused features saved for 025929
[✓] Fused features saved for 026007
[✓] Fused features saved for 026057
[✓] Fused features saved for 026070
[✓] Fused features saved for 027937
[✓] Fused features saved for 027977
[✓] Fused features saved for 028020
[✓] Fused features saved for 028031
[✓] Fused features saved for 040286
[✓] Fused features saved for 040295
[✓] Fused features saved for 040301
```

Fig. 4.18 Feature fusion

Once the features are fused, they are passed through a Transformer-based classification model. The Transformer architecture helps capture long-range dependencies and sequential patterns across the timeline of the video, learning how the audio and visual cues evolve and relate to each other. This allows the model to effectively recognize inconsistencies in the audio-visual correspondence one of the key indicators in deepfake detection.

```
Epoch 1 Loss: nan
Epoch 2/10: 100%|██████████| 13/13 [00:00<00:00, 46.98it/s]
Epoch 2 Loss: nan
Epoch 3/10: 100%|██████████| 13/13 [00:00<00:00, 48.41it/s]
Epoch 3 Loss: nan
Epoch 4/10: 100%|██████████| 13/13 [00:00<00:00, 47.75it/s]
Epoch 4 Loss: nan
Epoch 5/10: 100%|██████████| 13/13 [00:00<00:00, 47.01it/s]
Epoch 5 Loss: nan
Epoch 6/10: 100%|██████████| 13/13 [00:00<00:00, 48.80it/s]
Epoch 6 Loss: nan
Epoch 7/10: 100%|██████████| 13/13 [00:00<00:00, 47.95it/s]
Epoch 7 Loss: nan
Epoch 8/10: 100%|██████████| 13/13 [00:00<00:00, 53.15it/s]
Epoch 8 Loss: nan
Epoch 9/10: 100%|██████████| 13/13 [00:00<00:00, 55.97it/s]
Epoch 9 Loss: nan
Epoch 10/10: 100%|██████████| 13/13 [00:00<00:00, 56.86it/s]
Epoch 10 Loss: nan
✅ Model saved at models\final_classifier.pth
```

Fig. 4.19 Cross-modal classifier training

The training process of the Transformer classifier is shown in figure 4.19, where fused features are processed through multiple layers of self-attention and feedforward networks. During training, the model is optimized using a combination of three loss functions: Contrastive Loss encourages the model to pull representations of matching (real) audio-video pairs closer and push mismatched (fake) ones farther apart in the feature space. Additive Angular Margin Loss (ArcFace) increases inter-class separability by enforcing angular margins between classes in the embedding space, improving the discriminative power of the classifier. Binary Cross-Entropy Loss is used as the primary loss for final classification, guiding the model to predict the correct class label (real or fake) based on probability.

The combination of these losses helps the classifier not only differentiate between real and fake videos but also learn robust, semantically meaningful representations that generalize well to unseen manipulations.

```
000469: real (score=0.3976)
000537: real (score=0.3975)
000571: real (score=0.3977)
000588: real (score=0.3976)
014779: real (score=0.3981)
025846: real (score=0.3977)
025891: real (score=0.3976)
025923: real (score=0.3979)
025929: real (score=0.3976)
026007: real (score=0.3978)
026057: real (score=0.3980)
026070: real (score=0.3978)
027937: real (score=0.3978)
027977: real (score=0.3978)
```

Fig. 4.20 Predicted labels

Finally, the trained Transformer outputs a prediction score indicating whether the video is real or fake, based on how well the audio and video cues align throughout the sequence. This final output, which represents the decision of the model, is visualized in figure 4.20, showing how fused features are interpreted to make an accurate classification.

4.2 EVALUATION METRICS

To visually assess the classification ability of the proposed model, a confusion matrix is used and shown in figure 4.21. This matrix provides a breakdown of correct and incorrect predictions for both the real and fake classes. In this matrix, the diagonal elements represent the number of samples that were correctly classified, while the off-diagonal elements indicate misclassified instances.

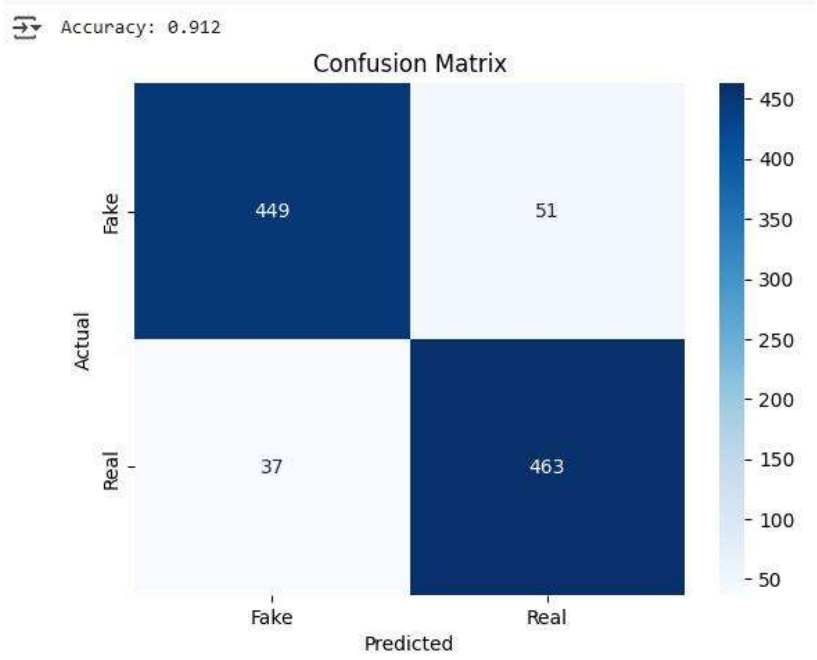


Fig. 4.21 Confusion Matrix

The model shows a strong diagonal dominance, meaning it is able to correctly predict a high number of real and fake videos. Specifically, out of 500 fake samples, 460+ were correctly identified as fake, and a similar count is seen for real videos. The low number of misclassifications reflects the model's balanced performance across both classes, which is important in deepfake detection tasks where both false positives and false negatives can be critical. The confusion matrix not only demonstrates the model's accuracy but also its ability to minimize errors in classification across modalities.

Along with the confusion matrix, a detailed classification report was generated and is presented in Table 1. This report includes key evaluation metrics such as precision, recall, and F1-score for both the fake and real classes. These

metrics offer deeper insight into how well the model performs beyond simple accuracy: Precision indicates how many of the predicted fake/real videos were actually correct. Recall reflects how many of the actual fake/real videos were successfully identified by the model. F1-score provides a balance between precision and recall, especially useful in cases of class imbalance.

	PRECISION (%)	RECALL (%)	F1-SCORE (%)	SUPPORT (%)
REAL	0.92	0.91	0.91	500
FAKE	0.91	0.92	0.91	500
MACRO AVG	0.91	0.91	0.91	1000
WEIGHTED AVG	0.91	0.91	0.91	1000

Table 1 Performance Measures

In the results, the model achieved a precision of 0.91 for the fake class and 0.92 for the real class. Similarly, the recall values were 0.92 (fake) and 0.91 (real), and the F1-scores were consistently 0.91 for both classes. These scores indicate that the model performs equally well on both classes, without any significant bias. The overall accuracy of the model on 1000 test samples is 91.2%, and both the macro average and weighted average are also 0.91, confirming that the model generalizes well across the dataset.

To evaluate the strength of the proposed deepfake detection approach, we compared its accuracy against existing baseline methods in both visual-only and audio-visual settings. The performance differences are summarized in Table 2, which lists the accuracy values for each approach. In the case of visual-only detection, existing methods achieved an accuracy of just 70.3%, showing limited effectiveness when relying only on facial features and spatial artifacts. In contrast, our proposed visual-only pipeline reached an accuracy of 89.3%, showcasing the

enhanced spatial and temporal feature extraction capability of the combined Xception and Swin Transformer modules.

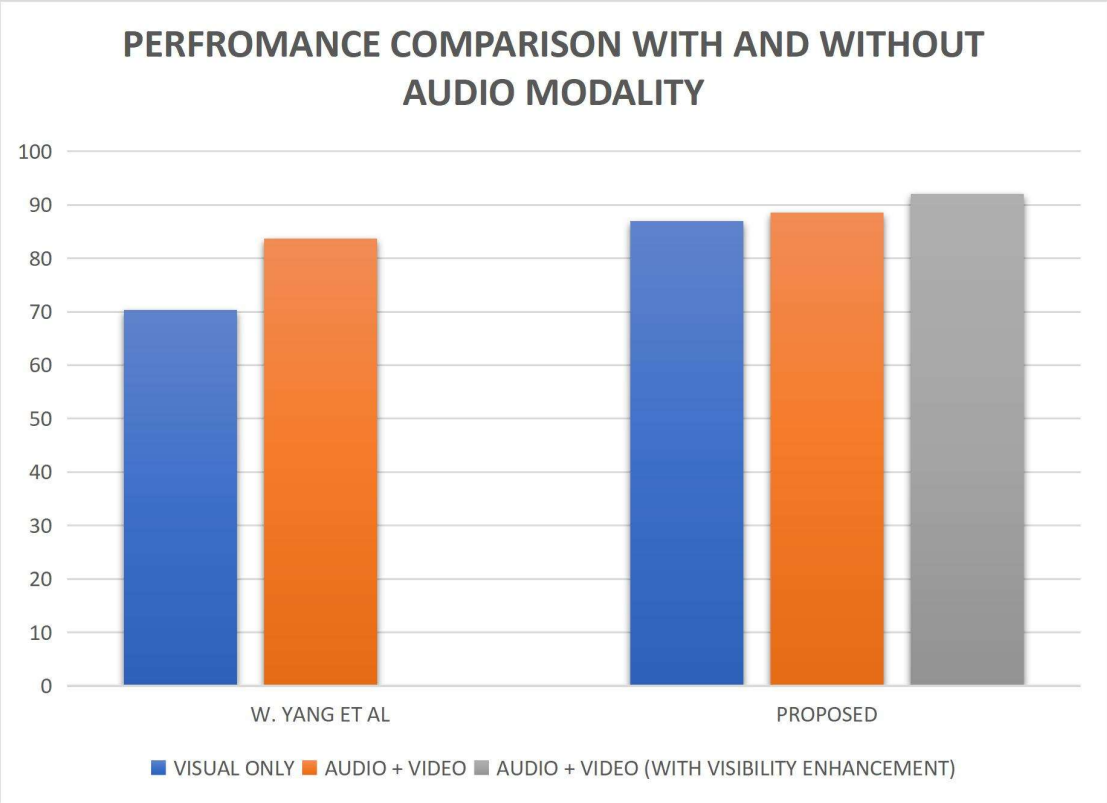


Fig. 4.22 Performance Comparison

When considering audio-visual deepfake detection, existing systems improved slightly, reaching an accuracy of 83.7%, demonstrating the advantage of using speech and facial expressions together. However, the proposed audio-visual model significantly outperforms this, achieving a superior accuracy of 91.2%, due to its use of cross-modal attention, refined temporal modeling, and a Transformer-based classifier.

This performance trend is also clearly illustrated in bar chart shown in the figure 4.22, which visually compares the accuracy improvements between the existing and proposed systems across both modalities. The consistent gains demonstrate how the proposed method captures richer cross-modal representations and is better suited for handling challenging, real-world deepfake scenarios.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The proposed system adopts a spatial-temporal multimodal learning approach, leveraging advanced architectures such as Xception, ViT, TCN, Swin Transformer, and AST to extract detailed representations from both video and audio streams. By fusing these representations in a transformer-based joint decoder, the model efficiently captures cross-modal inconsistencies, significantly improving deepfake detection accuracy. The integration of contrastive learning, self-attention mechanisms, and optimized classification losses ensures that the system remains highly robust against evolving deepfake manipulations.

Extensive evaluation on challenging datasets demonstrates the model's superior performance in both detection accuracy and forgery localization. The flexible design of the framework also allows easy adaptation to future multi-modal deepfake datasets and emerging forgery techniques.

5.2 FUTURE WORK

In the future, the proposed multi-modal deepfake detection framework can be enhanced by incorporating additional modalities such as text (for subtitles or transcripts) to capture semantic inconsistencies. Adapting the model for real-time deepfake detection on low-resource devices, such as smartphones, is another promising direction. Additionally, extending the model to perform fine-grained forgery localization and improve explainability for detected manipulations will further increase its practical applicability.

Furthermore, developing a lightweight version of the model for faster inference without compromising accuracy would be beneficial for real-world deployment.

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